Project

Planning

Requirements

- One main page, and 7/8 sub pages

This means creating the main page that will list all the articles and let you go to the add, view, update, and delete. Then the subpages are the Add, View, Update, and Delete pages + the student, tutor, and admin login pages, and maybe adding a homepage.

- Database that can hold the sample data

This means making a database that can hold all sample data and then putting said sample data in the database.

- Database has authentication for 3 groups: students, Tutors, and Administrators

This means making a collection that stores users into one of these 3 groups and then giving them permissions depending on what group they are a part of.

- Main page shows the articles names and options depending on permissions.

This means that when logged in as a student, you should only be able to view articles. When logged in as a tutor, you should be able to view, add, and update the articles. And when logged in as an admin, be able to view, add, update, and delete the articles.

- Make front end easier to use and look nice.

Add styling to the front end to make it easier to use and look nice visually.

Timeline of the 5 weeks

Week 1 get planning & analysis done

Week 2 do solution design and build the database

Week 3 fill the database with the sample data and start making front end

Week 4 finish making front end and start styling

Week 5 finish styling and all the documentation

Solution design

Architecture

Investigate possible database management systems appropriate to meet the requirements and select one. Justify your selection.

There are two main database systems that could meet the requirements being MySQL and MongoDB. The difference is that MySQL is a relational database and works well with multiple tables but needs detailed types for each field whereas MongoDB is a non-relational database and works great for databases that need no table relations and no strict defining of types for the fields.

I picked MongoDB because there is only one main collection, and it doesn't need to connect to any other collections, making it more suitable. Also, because it is easier to import the supplied sample data into MongoDB than MySQL, which is why I picked MongoDB.

Database Design

Design the data model. Identify main entities, attributes, data types, and relationships.

Articles collection:

Names	type	MYSQL (types)
_id	ObjectId	Int pri
Category	String	Varchar (250)
Type	String	Varchar (250)
Name	String	Varchar (250)
Born	String	Date
Died	String	Date
Nationality	String	Varchar (250)
Known For	String	Varchar (250)
Notable work	String	Varchar (250)
About	String	Varchar (500)
Year	String	Varchar (250) (can't be date because it has 2 dates: start and end)
Medium	String	Varchar (250)
Dimensions	String	Varchar (250)
Location	String	Varchar (250)
Designed by	String	Varchar (250)
Developer	String	Varchar (250)

Users collection:

Names	type	MYSQL (types)
_id	ObjectId	Int pri
Username	String	Varchar (250)
Password	String	Varchar (250)
Group	String	Varchar (250)

No relationships as all collections aren't connected

List appropriate queries as per the requirements and planning documents.

{ Category: "Mathematics" }

{ Category: "Arts" }

{ Category: "Technology" }

User Access Design

Advise on how the user security requirements should be implemented in regard to user authentication and authorization.

Firstly, for the user's security you will need to make sure that the data can keep confidentiality by using encryption, for example on the user's passwords so that they can't be seen. You would also need to make sure you use secure connections like http so that you can keep the integrity of the data. You will also need to make sure that the permissions are set up correctly so that the availability and authentication of data is only accessible by the correct users. You also need to make sure that users will be held accountable by making it you can trace back where the data entry originated from (within the bounds of the law). This will help keep the authentication and authorization of who can access the data secured.

There are also some requirements to make sure that the data is secure being:

- Make sure you have security requirements like this.
- Make sure the database can only be accessed through secure channels.
- Make sure all inputs are valid (like make it they can't change input settings in the input)

- Make sure that users need to be logged in to do actions that would need authorization.
- Make sure that the user groups can't change their permissions and can only do what their permissions say they can.

Define user roles and permissions.

roles	permissions
Students	View
Tutors	View, Add, Modify
Administrators	View, Add, Modify, Delete

User Experience Design

Suggest views required for the overall web application.

Article list view (when logged in as admin)

Login Student Login Tutor Logout of Admin

Add article

Filters:

Category: Technology V Name:

Name	Category	Actions
Java	Technology	View, Edit, Delete

Login Student

Login Tutor

Login Admin

Login as Student

[Return to main page]

Username:

Password:

Login Bypass

Logout of Student Login Tutor Login Admin

Current Articles

Filter:

Category: Technology **Name:**

Name	Category	Actions
Java	Technology	[view]

Login as Tutor

[Return to main page]

Username:

Password:

Login Bypass

Login Student Logout of Tutor Login Admin

Current Articles

[Make a article]

Filter:

Category: Technology **Name:**

Name	Category	Actions
Java	Technology	[view] [edit]

Login as Admin

[Return to main page]

Username:

Password:

Login Bypass

Login Student Login Tutor Logout of Admin

Current Articles

[Make a article]

Filter:

Category: Technology **Name:**

Name	Category	Actions
Java	Technology	[view] [edit] [delete]

View Article

[Return to the main page]

Article ID: 69c5d49caabf80825cb2c801

Category:
Technology
Type:
Programming Language
Name:
Java
Born:
Died:
Nationality:
Known For:
Notable Work:
About:
Java is a general-purpose computer-programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few implementation dependencies as possible. It is intended to let application developers "write once, run anywhere", meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM) regardless of computer architecture. As of 2016, Java is one of the most popular programming languages in use, particularly for client-server web applications, with a reported 9 million developers. Java was originally developed by James Gosling at Sun Microsystems (which has since been acquired by Oracle Corporation) and released in 1995 as a core component of Sun Microsystems' Java platform. The language derives much of its syntax from C and C++, but it has fewer low-level facilities than either of them.
Year:
Medium:
Dimensions:
Location:
Designed By:
James Gosling
Developer:
Sun Microsystems

Add a Article

[Return to the main page]

Category:
Type:
Name:
Born:
Died:
Nationality:
Known For:
Notable Work:
About:
Year:
Medium:
Dimensions:
Location:
Designed By:
Developer:

Add [Cancel]

Edit a Article

[Return to the main page]

Category: Technology
Type: Programming Language
Name: Java
Born:
Died:
Nationality:
Known For:
Notable Work:
About: Java is a general-purpose computer-programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few implementation dependencies as possible. It is intended to let application developers "write once, run anywhere", meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM)
Year:
Medium:
Dimensions:
Location:
Designed By: James Gosling
Developer: Sun Microsystems

Update [Cancel]

Delete Article

[Return to the main page]

Article ID: 69c5d49caabf80825cb2c801

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Are you sure you want to delete this Article?

Delete Cancel

GitHub Link

https://github.com/lordsepticplier/SchoolHub_DataManagement_Project